

LUCAS GONÇALVES

AI Engineer | RAG & LLM Systems · Agent Orchestration · Evaluation

Brazil (GMT-3) · Remote-ready, any time zone · Open to relocation

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SUMMARY

AI Engineer with 2+ years building retrieval-augmented and agentic LLM systems in Python with FastAPI, LangGraph, Qdrant and pgvector. Built a self-hostable RAG service that ships a semantic-entropy confidence score with every answer, calibrating its agent domain gate to 96.7% true-positive at 3.3% false-positive over a 120-question labeled sweep. Drove the retrieval and grounding core of an AI advising platform to 1,149 automated tests, and built a multi-agent code-generation service behind a deterministic release gate. Targeting AI Engineer roles in RAG, agent orchestration and applied LLM systems.

KEY ACHIEVEMENTS

- **Calibrated an LLM agent's domain gate from measurement rather than intuition:** an ROC sweep over 120 labeled questions set it at 96.7% true-positive / 3.3% false-positive, frozen as a fixture so CI fails if a default drifts.
- **Cut an incorrect field-parts return rate from ~20% in 2025 to 0% by June 2026,** against a 6% Six Sigma target, removing recurring freight cost from unnecessary returns across a 150+ dealership network.
- **Improved sentiment classification to 0.887 macro F1 against a 0.584 frozen-features baseline,** a 51.9% relative gain concentrated in the rarest classes, driven by per-class error analysis.

EXPERIENCE

Unipac | Software Engineer - AI & Data | 09/2026 - Present | Pompeia, SP, Brazil (Hybrid)

Packaging and polymer processing manufacturer. Internal AI platform for automated software generation.

- Architected a seven-phase specialist agent pipeline under a coordinator that delegates rather than writes code, generating 20+ full-stack applications end to end, each submitted to 13 deterministic verifiers before release. (*LangGraph, LangChain, deepagents, FastAPI*)
- Eliminated model self-approval from the release path by deriving the security verdict from scanner receipts produced in a short-lived offline micro-VM and bound to the SHA-256 of the delivered tree, applied by a pure function the pipeline cannot bypass, locked by 32 contract tests.
- Designed a seven-layer runaway-containment stack (session cap, tool cap, delegation cap, idempotent loop breaker, retry, fallback, circuit breaker), each layer built in response to an observed incident, including a tool invoked 72 times in a loop and a job stalled 30 minutes by budget-resetting re-delegation.

PM Accelerator | AI Engineer | 06/2026 - 08/2026 | Miami, FL (Remote)

Early-stage US ed-tech startup. AI-powered college-advising platform.

- Drove the retrieval and grounding core, engineering a hybrid retrieval engine from scratch with no RAG framework over a corpus of ~1,800 US institutions: exact name and alias matches ranked above pgvector cosine similarity on a deliberate no-score-fusion design, behind an HNSW index, returning an explicit insufficient-data signal below threshold.
- Eliminated a class of LLM hallucination structurally rather than by prompt instruction: a refuse-before-call gate that spends neither an API call nor quota, and a verifier that flags figures no retrieved record supports, catching one institution's statistic quoted under another's name.
- Scaled the backend suite to 1,149 automated tests at a test-to-code ratio above 1.7:1, and cut CI dependency footprint ~9x (800MB to 90MB) by serving the same embedding model through an ONNX runtime instead of a full deep-learning framework.
- Delivered 19 of 29 tracked board items on a weekly cadence against written specifications, coordinating asynchronously across time zones.

Jacto | After-Sales - Data & Automation | 11/2024 - 06/2026 | Pompeia, SP, Brazil (Hybrid)

Global agricultural machinery manufacturer, 3,000+ employees, operating in 100+ countries.

- Built an automated data-collection pipeline in Python (Selenium, BeautifulSoup) that scrapes hundreds of fiscal address records from a foreign government registry, translates them and organizes them by geographic division, removing up to 5 hours a day of manual collection and formatting and feeding a Salesforce integration with load-ready files.
- Cut the incorrect field-parts return rate from ~20% in 2025 to under 5% in 2026, closing June 2026 at 0% against a 6% Six Sigma target, removing recurring freight costs from unnecessary returns.
- Audited and reconciled the field-parts return flow across 150+ master dealerships, tracking millions of BRL in parts movement through on-site and remote audits.

CIAG | Software Engineer - Synapse Program | 02/2024 - 10/2024 | Pompeia, SP, Brazil (Hybrid)

Regional innovation program. Internal LLM assistant for a domain-specific knowledge base.

- Led development of a natural-language assistant on an LLM (Python, FastAPI, OpenAI API) with contextual RAG, designing the REST API layer between the application and the model and keeping answers grounded in a region-specific knowledge base.

PROJECTS

SmartB100: Self-hostable RAG API with semantic-entropy hallucination scoring

Python · FastAPI · Qdrant · Ollama · LangGraph · Docker · github.com/LukeSantossz/sb100_agents

- Architected a self-hostable RAG service on FastAPI, Qdrant and Ollama, with every significant trade-off recorded in 15 Architecture Decision Records that name and reject the alternatives considered.
- Implemented semantic-entropy hallucination scoring (Farquhar et al., 2023): N-sample generation, embedding-similarity clustering with a cache reducing the pass from $O(N^2)$ to N embed calls, and normalized Shannon entropy, behind a gate that degrades to a neutral score rather than failing.
- Reached 372 automated tests at 89.8% branch coverage of the core packages, with CI running ruff, mypy --strict, pytest and a Docker check asserting the build toolchain is absent from the runtime image.

tweet-sentiment-analysis: NLP fine-tuning pipeline with Rust preprocessing

Python · PyTorch · HuggingFace Transformers · Rust · Polars · github.com/LukeSantossz/tweet-sentiment-analysis

- Delivered 0.887 macro F1 / 92.3% accuracy against a frozen-features linear-probe baseline at 0.584 / 67.5%, a 51.9% relative gain concentrated in the rarest classes, after per-class error analysis drove a pivot in task and backbone.
- Rewrote preprocessing as a Rust CLI benchmarked at 28.5x the Python reference on 1M rows (33.08s to 1.16s), with output parity validated at three scales.

Open source

- **my-framework** (MIT), github.com/LukeSantossz/my-framework: engineering-standards framework whose pre-push cross-provider review gate is a pluggable adapter chain telling "reviewed" from "reviewer unavailable". 101 shell tests; reused across five repositories.
- **AutoMQ**, github.com/AutoMQ/automq PR #3261: accepted fix in a Kafka-compatible streaming platform: response-size validation so a count mismatch fails pending requests cleanly instead of raising IndexOutOfBoundsException. 157 lines of tests for a 6-line fix.

SKILLS

Languages: Python, SQL, TypeScript, Rust

LLM, RAG & Agents: RAG, LangGraph, LangChain, agent orchestration, LLM-as-judge evaluation, semantic entropy, embeddings, Anthropic API, OpenAI API

ML: PyTorch, HuggingFace Transformers, scikit-learn, fine-tuning

Backend & Data: FastAPI, PostgreSQL, pgvector, Qdrant

Cloud & Practices: Docker, GitHub Actions (CI/CD), pytest, Test-Driven Development

PUBLICATIONS & AWARDS

- "Soil Texture Classification by Image: Deep Feature Learning vs. Handcrafted Methods for Precision Agriculture", poster, 17th ICPA / 11th ConBAP, Porto Alegre, July 2026.
- **3rd place, 16th FETEPS 2025** (Centro Paula Souza Technology Fair), among 1,300+ entries.
- **FAPESP grant 2024/00985-1**, volunteer researcher on the funded project behind SmartB100.

EDUCATION

Fatec Shunji Nishimura | Technology Degree (CST), Big Data in Agribusiness | 01/2024 - 11/2026

Languages: English (B2, Upper Intermediate) · Portuguese (Native)